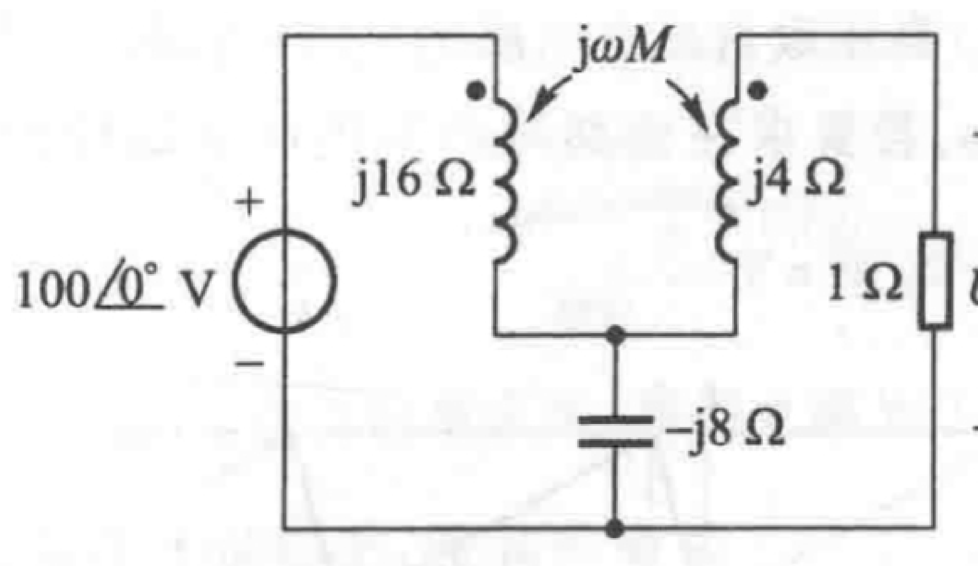
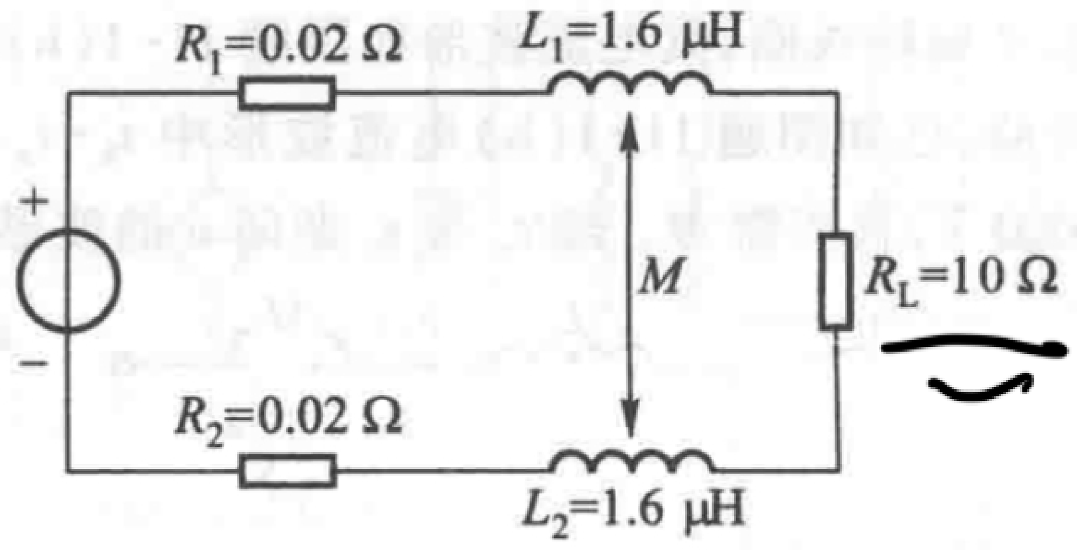
第十一章作业



图题 11-4



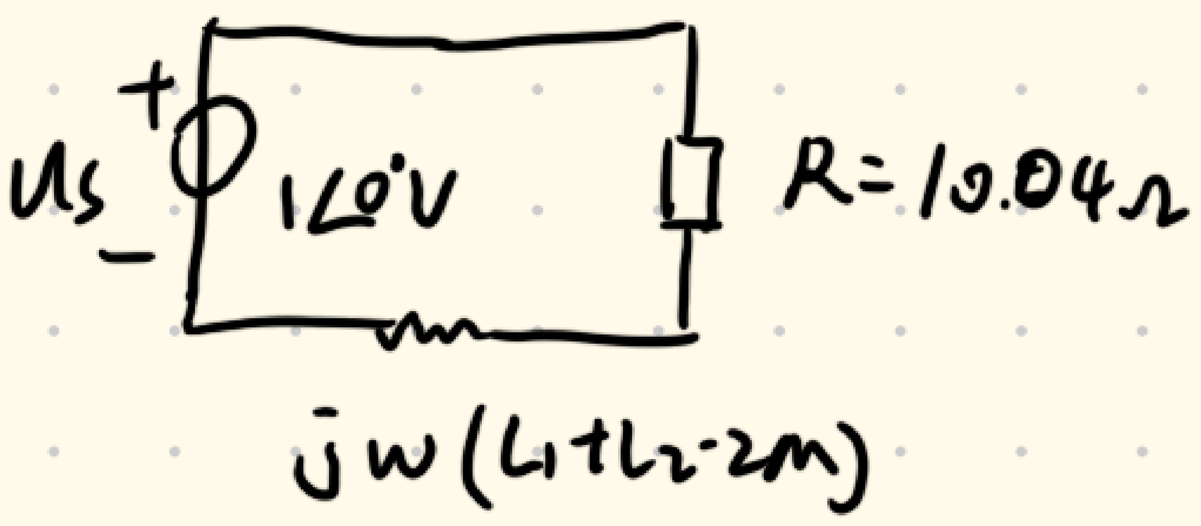
图题 11-5

11-5 同轴电缆的外导体与内导体之间总存在一些互感，图题 11-5 所示电缆用来传送 1 MHz 信号至负载 R_L，试计算耦合系数 k 为 0.75 和 1 时传送给负载的功率。已知 $\dot U_s = 1\angle 0°$ V。

$\frac{M}{\sqrt{L_1L_2}} = k \Rightarrow M = k\sqrt{L_1L_2}$.

① 当 k=1 时 $M=1.6\ \mu H$

② 当 k=0.75 时 $M=1.2\ \mu H$



① 当 k=1 时. $L_1+L_2-2M=0$

$\therefore \dot I = \frac{\dot U}{R} = \frac{1\angle 0° V}{10.04\Omega} = 0.0996\angle 0° A$.

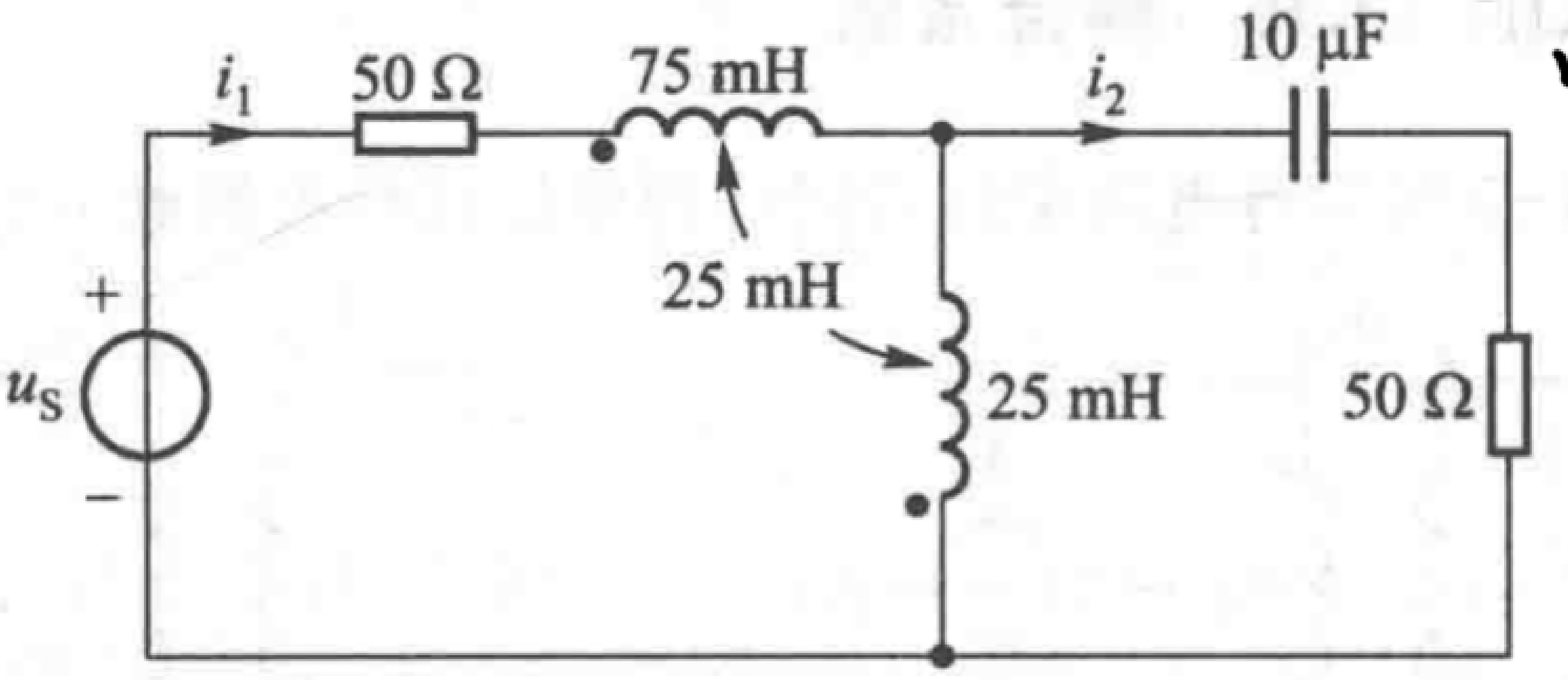
$P = I^2R_L = 99.2\ mW$

② 当 k=0.75 时. $L = L_1+L_2-2M = 0.8\ \mu H$

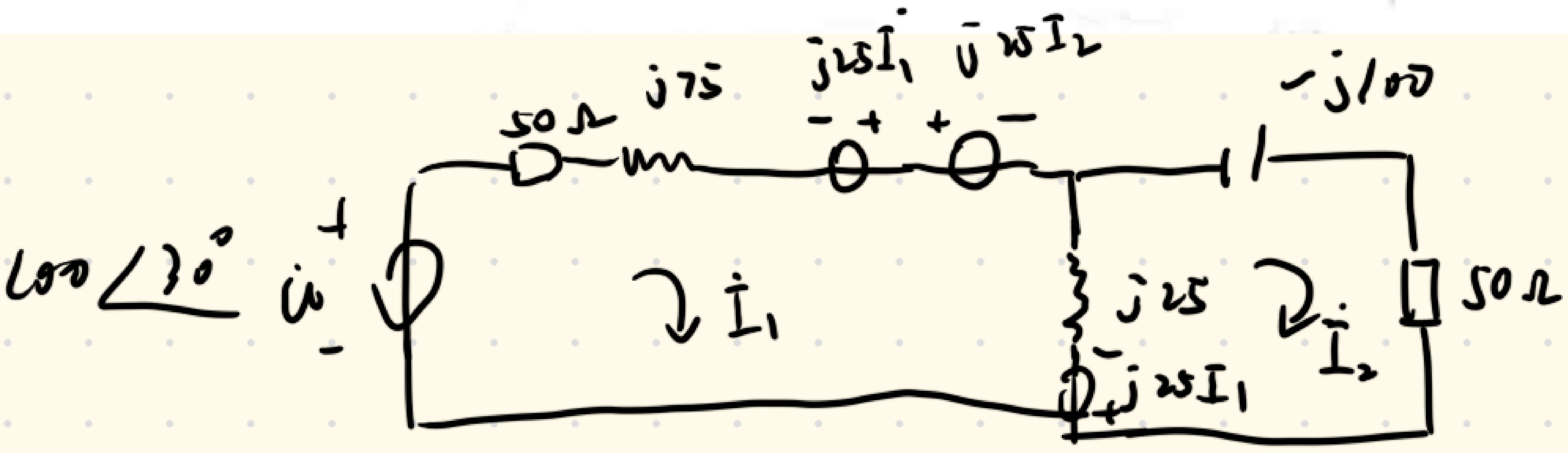
$\dot I = \frac{\dot U}{j\omega L+R} = \frac{1\angle 0°}{j2\pi\times10^6\times0.8\times10^{-6}+10.04} = \frac{1\angle 0°}{j1.6\pi+10.04} = \frac{1\angle 0°}{11.23\angle 26.59} = 0.089\angle -26.59°$

$P = I^2R_L = 0.089^2\times 10 = 79.2\ mW$

11-6 求图题 11-6 所示电路中的 i_1 和 i_2，已知 $u_S(t) = 100\cos(10^3t+30°)$ V。 $w=10^3$



图题 11-6



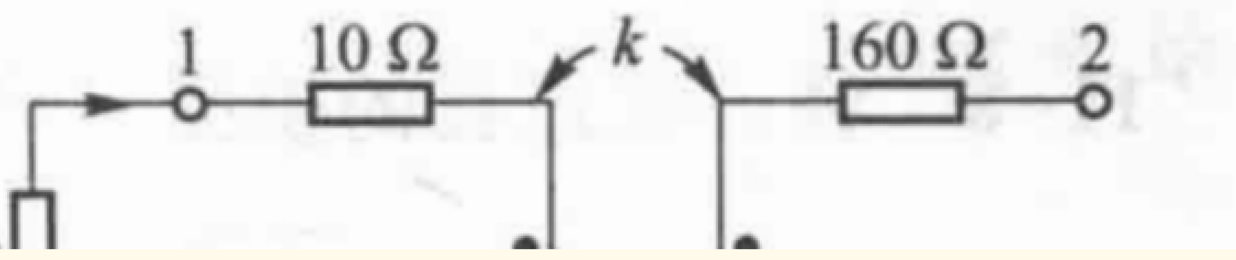
$$\begin{cases}(50+j75)\dot I_1 - j25\dot I_1 + j25\dot I_2 + j25(\dot I_1-\dot I_2) - j25\dot I_1 - \dot U = 0\\ -j100\dot I_2 + 50\dot I_2 + j25\dot I_1 + j25(\dot I_2-\dot I_1) = 0\end{cases}$$

$$\begin{cases}(50+j50)\dot I_1 - \dot U = 0\\ (-j75+50)\dot I_2 = 0\end{cases} \Rightarrow \begin{cases}\dot I_1 = \dfrac{100\angle 30^\circ}{50\sqrt2\angle 45^\circ} = \sqrt2\angle -15^\circ\\ \dot I_2 = 0\end{cases}$$

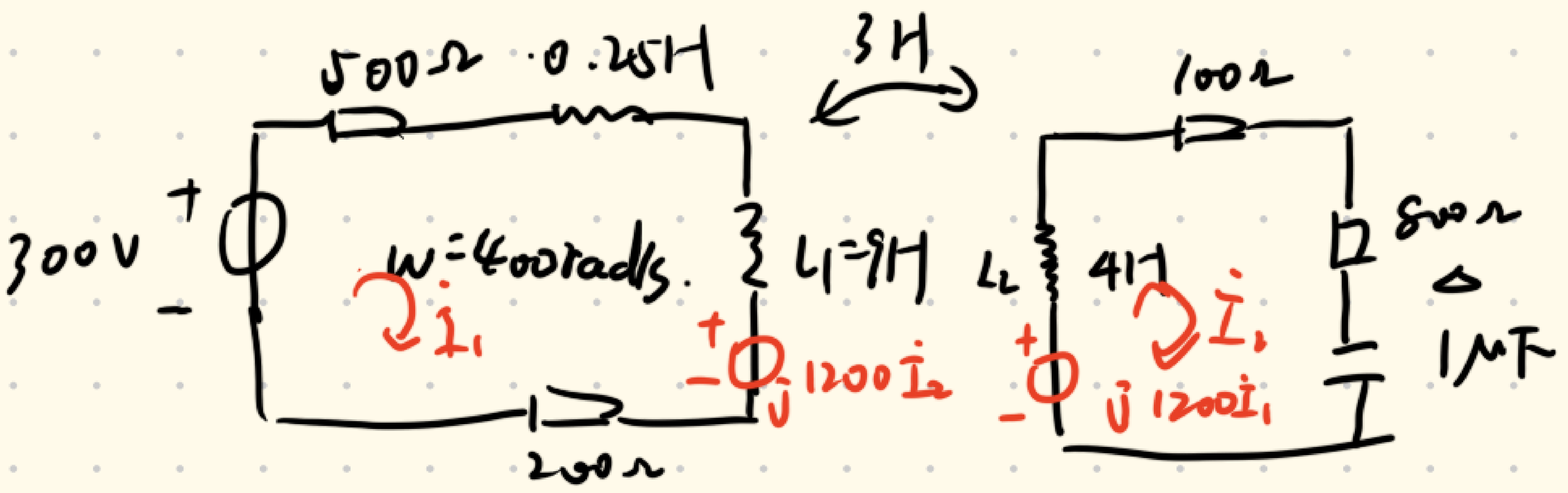
$\therefore\ i_1(t) = \sqrt2\cos(10^3t-15^\circ)$ A

$i_2(t) = 0$ A.

11-9 已知空心变压器的参数：$L_1=9$ H、$R_1=200\ \Omega$、$L_2=4$ H、$R_2=100\ \Omega$ 及 $k=0.5$。所接负载为 800 Ω 电阻和 1 μF 电容串联，所接正弦电压源频率为 400 rad/s，电压有效值为300 V，内阻为 500 Ω，内电感为 0.25 H。试求传送给负载的功率 P 和空心变压器的功率传输效率。



$M = \sqrt{L_1L_2}\cdot k = 3H$



$-j\dfrac{1}{1\times10^{-6}\times400}$

$=-j\frac14\times10^4$

$=$

回路1：$(500+200+j100+j3600)\dot I_1 - j1200\dot I_2 = 300\angle 0^\circ$

回路2：$(900+j1600-j2500)\dot I_2 - j1200\dot I_1 = 0$

$37\times3=111$ $-132-48$

$$\Rightarrow\begin{cases}(700+j3700)\dot I_1 - j1200\dot I_2 = 300\angle0^\circ\\ (900-j900)\dot I_2 - j1200\dot I_1 = 0\end{cases}$$

$$\Rightarrow\begin{cases}(7+j37)\dot I_1 = 3+j12\dot I_2\\ j4\dot I_1 = (3-j3)\dot I_2\end{cases}$$

$\dot I_1 = \dfrac{3-j3}{j4}\dot I_2 = \dfrac34(-1-j)\dot I_2$

$\Rightarrow \dfrac{3(7+j37)(-1-j)}{4}\dot I_2 = 3+j12\dot I_2$

$3(7+j37)(-1-j)\dot I_2 = 12+j48\dot I_2$

$(21+j111)(-1-j)\dot I_2 - j48\dot I_2 = 12$

$(-21+111-j111-j21-j48)\dot I_2 = 12$

$(90-j180)\dot I_2 = 12$

$\dot I_2 = 0.0596\angle 63.43^\circ$

$P_2 = (0.0596)^2\cdot 800 \approx 2.842$ W

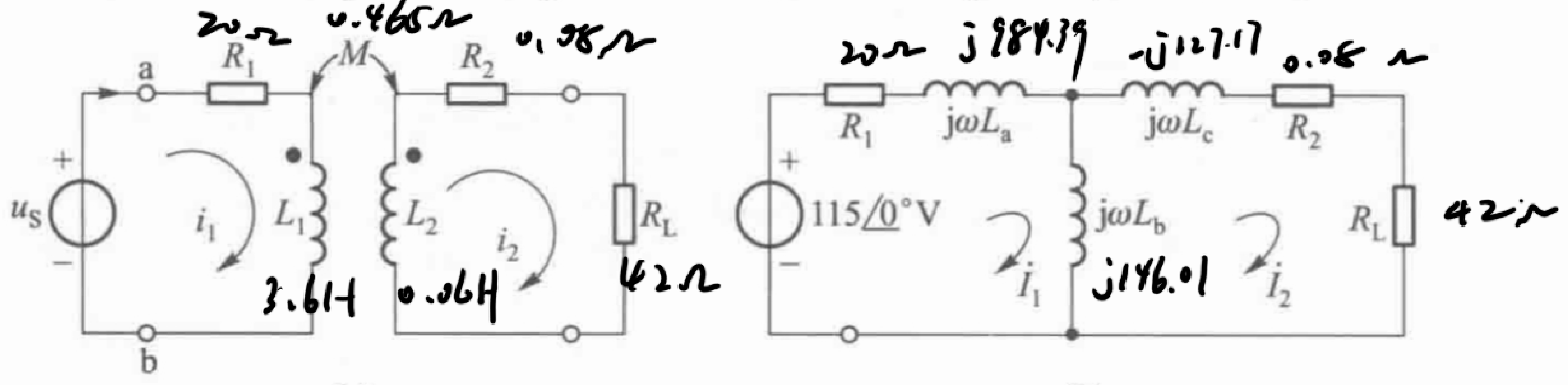
$$\dot{I}_1 = \frac{3}{4}(-j)\cdot\frac{12}{90-j180} = 0.0632\angle{-71.57^\circ}$$

$$P_1 = U\dot{I}_1\cos\theta = 300\times0.0632\times\cos(-71.57) \approx 6.00\,W$$

$$\eta = \frac{P_2}{P_1} = \frac{2.842}{6-500\times0.0632^2} \approx 71\%$$

11-11 试利用 T 形去耦电路求解例 11-5。

解 为方便计，特将例 11-5 电路图绘为图题解 11-11(a)，其 $L_1=3.6$ H、$L_2=0.06$ H、$M=0.465$ H、$R_1=20$ Ω、$R_2=0.08$ Ω、$R_L=42$ Ω、$u_S=115\sqrt{2}\cos(314t)$ V。



(a) (b)

$$L_a = L_1 - M = 3.6 - 0.465 = 3.135\,H$$
$$L_c = L_2 - M = -0.405\,H$$
$$L_b = M = 0.465\,H$$

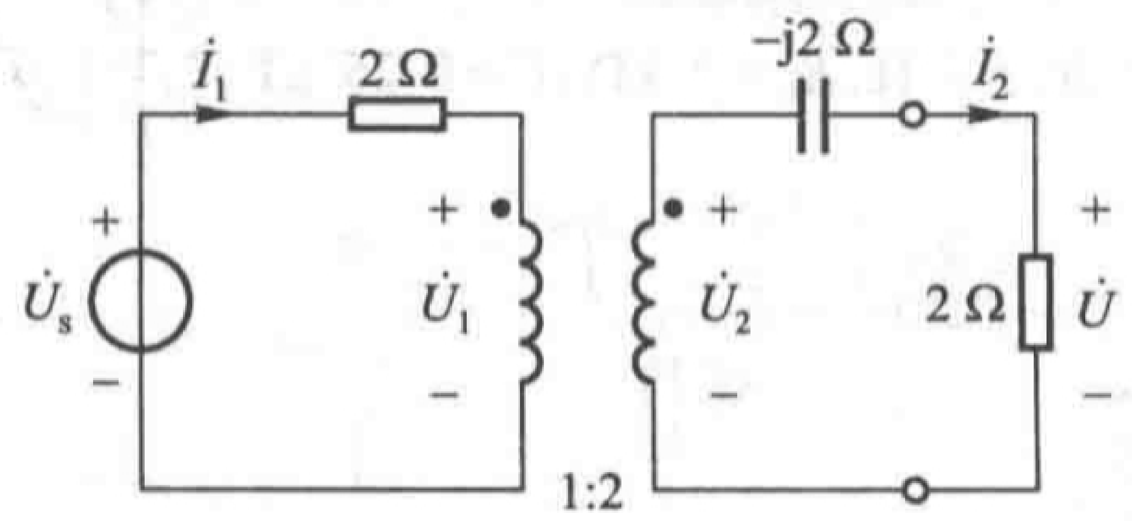
网孔1： $(20+j1130.4)\dot{I}_1 - j146.01\dot{I}_2 = 115\angle 0^\circ$

网孔2： $(42.08+j18.84)\dot{I}_2 - j146.01\dot{I}_1 = 0$

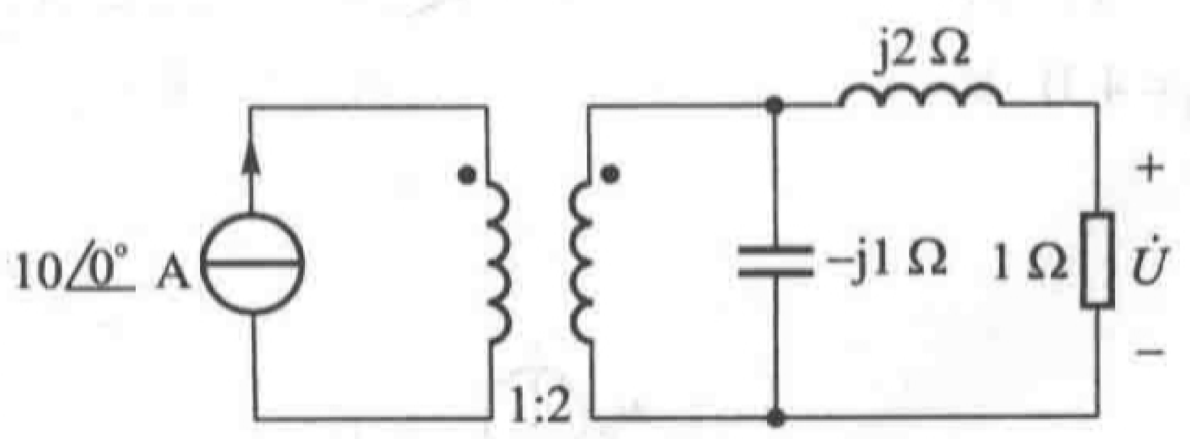
$$(20+j1130.4)\dot{I}_1 - j146.01\cdot\frac{j146.01}{42.08+j18.84}\dot{I}_1 = 115$$

$$\dot{I}_1 = 0.1106\angle{-64.85^\circ}\,A = 110.6\angle{-64.85^\circ}\,mA$$

11-13 电路如图题 11-10 所示，已知 $\dot{U}=10\angle 0^\circ$ V，试求 $\dot{U}_s$。



图题 11-10



图题 11-11

$$\dot{I}_2 = \frac{\dot{U}}{2} = 5\angle 0^\circ\,A$$

$U_c = \dot I_2 \cdot Z_c = 5\angle 0° \cdot 2\angle -90° = 10\angle -90°\ V$

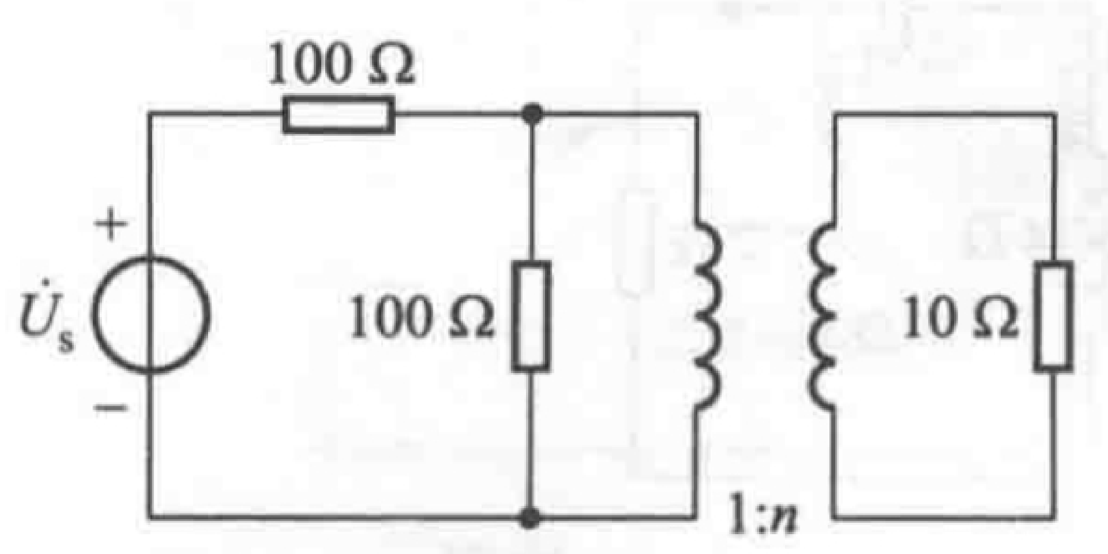
$\therefore\ \dot U_2 = \dot U_c + \dot U = (10 - j10)\ V$

$\therefore\ \dot U_1 = \frac{1}{2}\dot U_2 = (5 - j5)\ V$

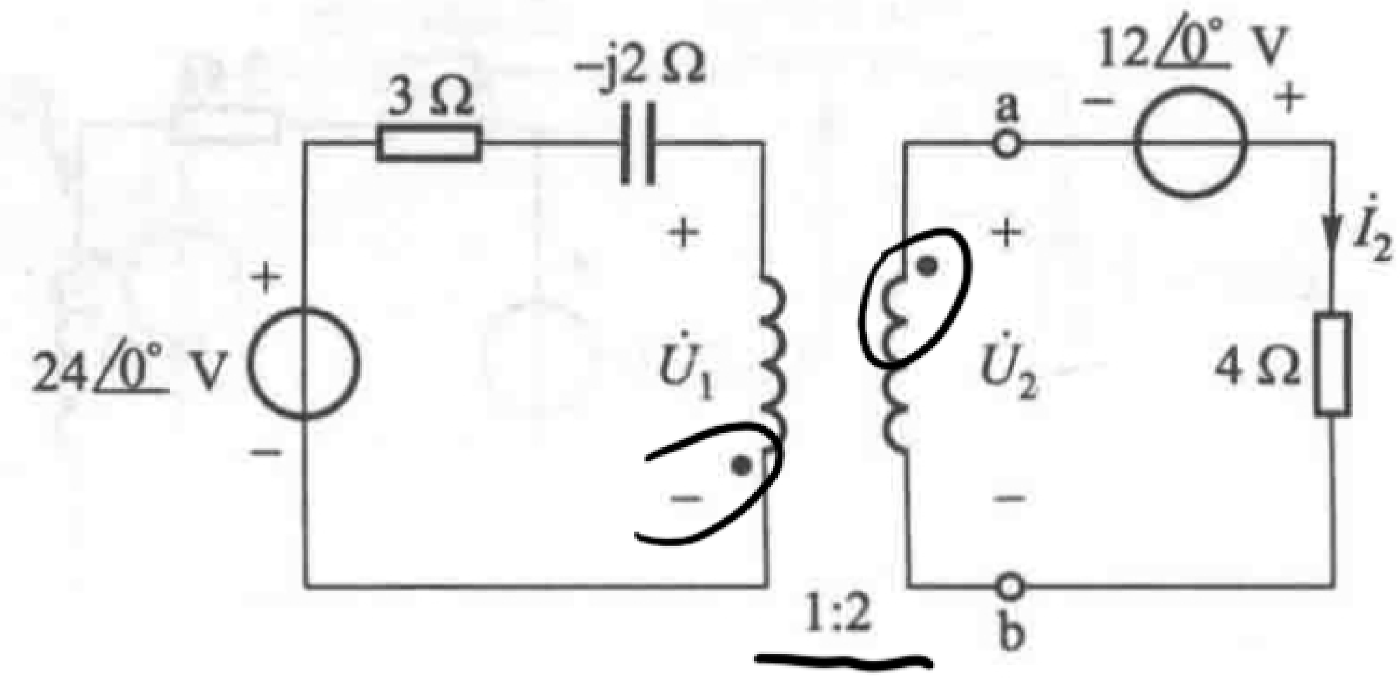
$\dot I_1 = 2\dot I_2 = 10\angle 0°\ A$

$\therefore\ \dot U_s = \dot U_1 + 2\dot I_1 = 20 + 5 - j5 = (25 - j5)\ V$

$= 25.495\angle -11.31°\ V$



图题 11-12

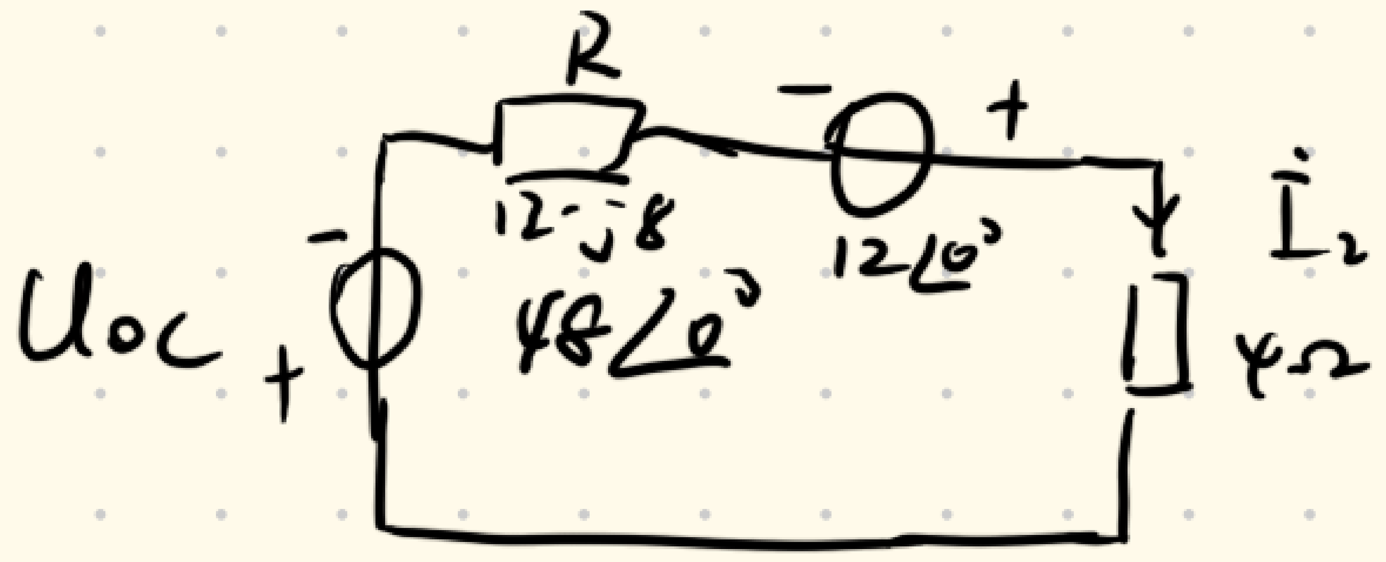


图题 11-13

11-16　电路如图题 11-13 所示，试求 ab 左侧的戴维南等效电路，并由此求解 $\dot I_2$。

$R = 2^2(3 - j2) = (12 - j8)\ \Omega$

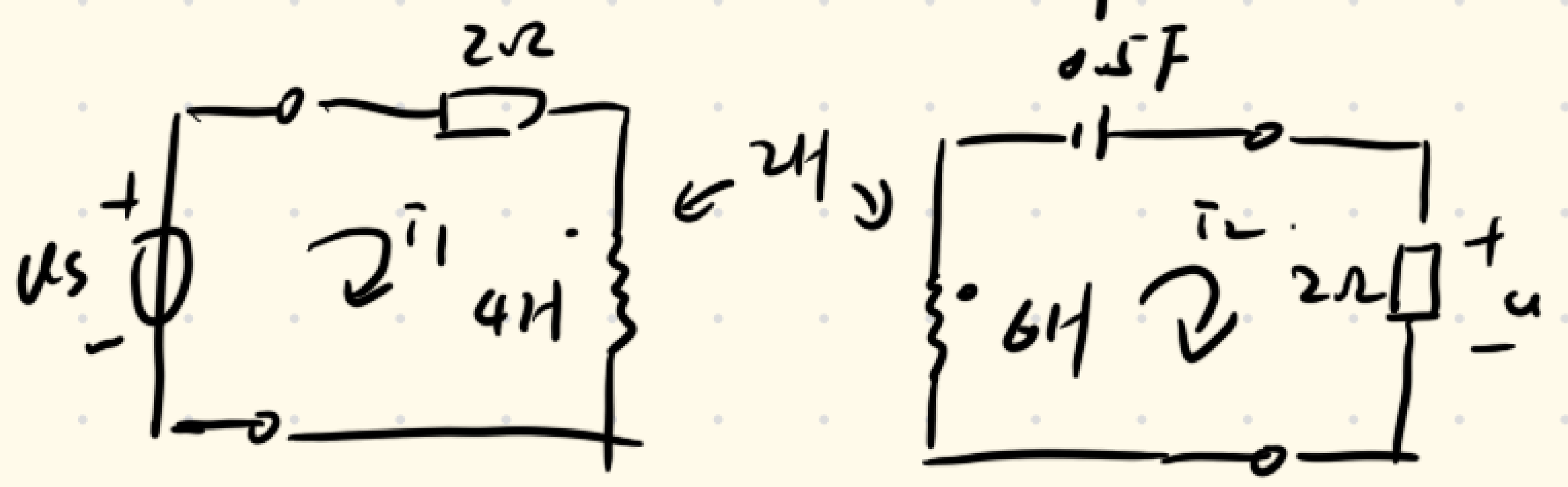
$U_{oc} = -2 \cdot 24\angle 0°\ V = -48\angle 0°\ V$



$(12 - j8 + 4)\dot I_2 - 12\angle 0° + 48\angle 0° = 0$

$$\dot I_2 = \frac{-36\angle 0°}{16 - j8} = -2.01\angle 26.57°\ A$$

11-3　$u_s = 24\cos(t + 30°)V$，试求输出电压 $u(t)$



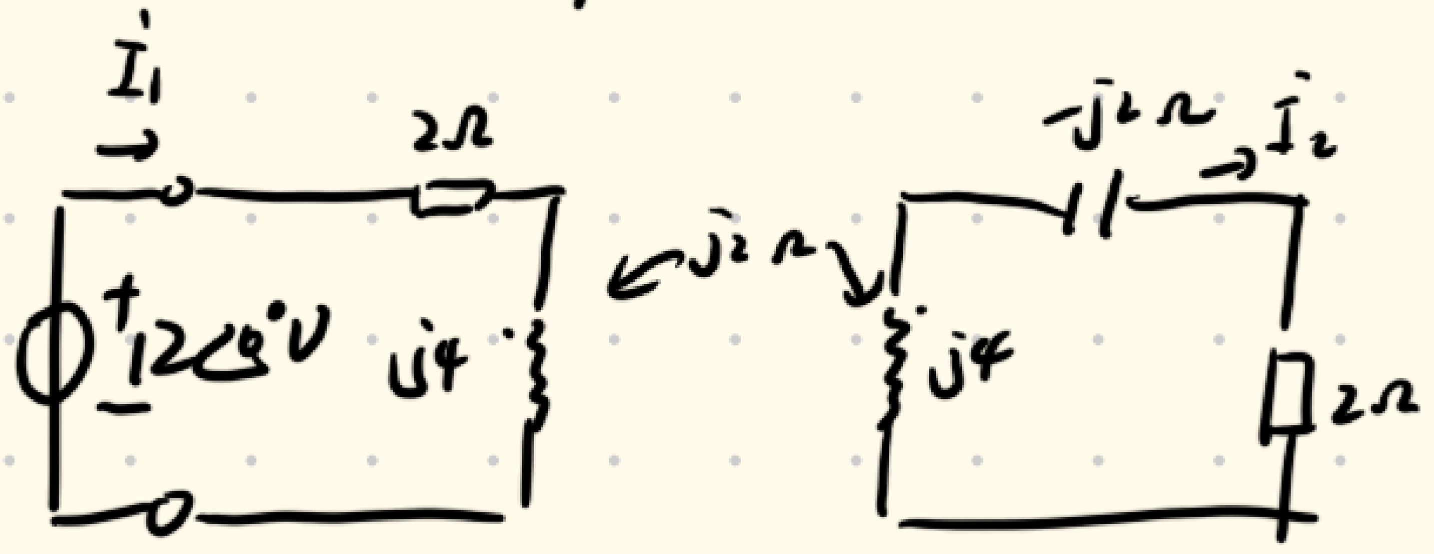
$\dot U_{sm} = 24\angle 30°$

$Z_1 = 2 + j4 + \frac{\omega^2 M^2}{Z_{22}} = 2 + j4 + \frac{2}{1 + j2}$　　$\therefore\ \dot I_{1m} = \frac{24\angle 30°}{2 + j4 + \frac{2}{1 + j2}} = 6\angle 156.8°\ A$

$$\dot{U}_m = j2\dot{I}_{1m} \times \frac{2}{2-j2+j6} = 5.36\angle 3.4^\circ\ V$$

$$u(t) = 5.36\cos(t+3.4^\circ)\ V$$

11-8 求对电源端口的输入阻抗，电流 $\dot{I}_1$ 和 $\dot{I}_2$

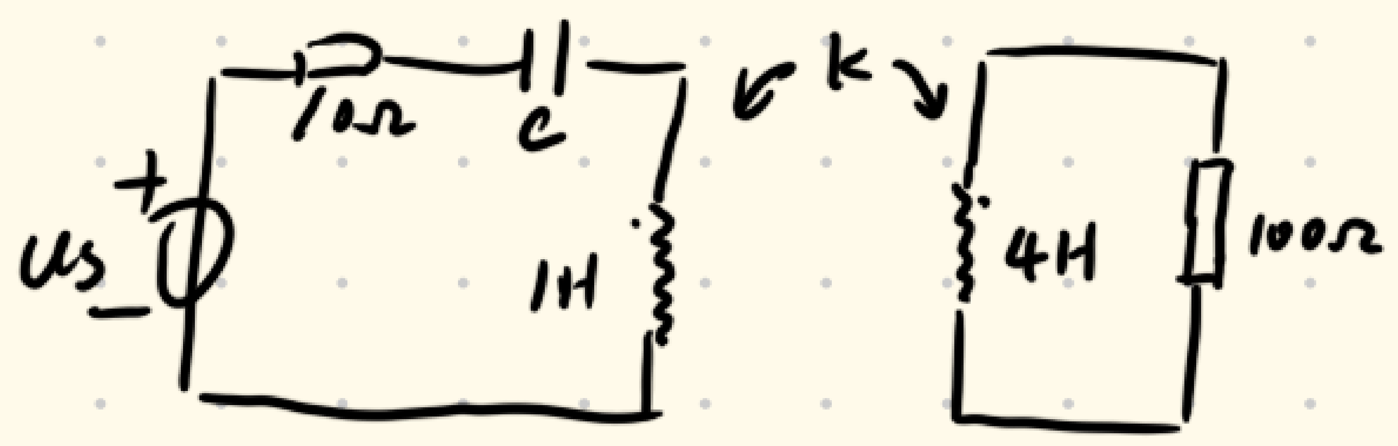


$$Z_1 = \frac{2^2}{j2+2} + 2 + j4 = 3+j3\ \Omega$$

$$\dot{I}_1 = \frac{12\angle 0^\circ}{3+j3} = 2\sqrt{2}\angle -45^\circ A$$

$$\dot{I}_2 = \frac{j2\dot{I}_1}{2-j2+j4} = 2\angle 0^\circ A$$

11-21 $\omega=1000rad/s$ 时串联谐振，试求C，已知 $k=0.5$



$$M = k\sqrt{L_1L_2} = 0.5\sqrt{4} = 1\ H$$

$$Z_{ref} = \frac{\omega^2 M^2}{Z_{22}} = \frac{10^6}{100+j4000} = 6.25 - j249.8\ \Omega$$

$$Z_{in} = Z_{11} + Z_{ref} = 10 - j\frac{1}{1000C} + j1000 + 6.25 - j249.8$$

虚部为零 $\Rightarrow -\frac{1}{1000C} + 1000 - 249.8 = 0 \Rightarrow C = 1.33\mu F$